(i)

$$y = x^3 - 3qx - q(1+q)$$

$$\frac{dy}{dx} = 3x^2 - 3q = 3(x^2 - q)$$

$$\frac{dy}{dx} = 0 \implies x = \pm\sqrt{q}$$

When $x = -\sqrt{q}$,

$$y = -q\sqrt{q} + 3q\sqrt{q} - q - q^2 = -q(\sqrt{q} - 1)^2 < 0$$

When $x = \sqrt{q}$,

$$y = q\sqrt{q} - 3q\sqrt{q} - q - q^2 = -q(\sqrt{q} + 1)^2 < 0$$

$\because$ Both turning points are below the x-axis.

$$x \to -\infty \implies y \to -\infty$$
$$x \to \infty \implies y \to \infty$$

$\therefore$ the curve crosses the x-axis once only

(ii)

$$x = u + \frac{q}{u}$$

$$\begin{aligned} x^3 - 3qx - q(1+q) &= 0 \\ \left(u + \frac{q}{u}\right)^3 - 3q\left(u + \frac{q}{u}\right) &= q(1+q) \\ u^3 + \frac{q^3}{u^3} &= q(1+q) \end{aligned}$$

Let $t = u^3$

$$t^2 - q(1+q)t + q^3 = 0$$

$$(t - q)(t - q^2) = 0$$

$$u^3 = q \quad \text{or} \quad u^3 = q^2$$

$$u = \sqrt[3]{q} \quad \text{or} \quad u = \sqrt[3]{q^2}$$

$$x = u + \frac{q}{u} = \sqrt[3]{q} + \sqrt[3]{q^2}$$

(iii)

$$\alpha + \beta = p,\ \alpha\beta = q$$

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = p^3 - 3qp$$

$$\begin{aligned}(\alpha^2 - \beta)(\beta^2 - \alpha) &= 0\\ \alpha^2\beta^2 - (\alpha^3 + \beta^3) + q &= 0\\ q^2 - (p^3 - 3qp) + q &= 0\\ p^3 - 3qp - q(1 + q) &= 0\end{aligned}$$

By **(ii)**, the real root is

$$p = \sqrt[3]{q} + \sqrt[3]{q^2}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: July 2, 2026